

Modeling Loci and Intersections with Conic Sections

Answer Key

Algebra 2 — Conics and Modeling

Quick Answers

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|---|---|
| 1. 25 | 8. $x = 2, y = 1$ |
| 2. -4, 4 | 9. -10, 10 |
| 3. 3 | 10. $x = 3, y = 4, x = 4, y = 3, x = -3, y = -4,$
$x = -4, y = -3$ |
| 4. -12, 12 | 11. -2, 4 |
| 5. $x = 3, y = 4, x = -4, y = -3$ | 12. $x = 4, y = 2, x = -4, y = 2, 4, 4$ |
| 6. 3, -4, 6 | |
| 7. $x = 3, y = 4, x = -3, y = 4, x = 0, y = -5$ | |
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Common wrong answers

1. *If they got 101: added the center coordinates instead of subtracting them, so the translation went the wrong way*
3. *If they got 12: divided by the focal distance p instead of by $4p$ when writing the parabola from its focus and directrix*
6. *If they got 3.31662: read the constant term as the radius squared and never added back the two completing-the-square constants*
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1. Is $(7, 1)$ on the circle with center $(3, -2)$, radius 5?

The center-radius form subtracts the center coordinates, and a k of -2 makes that subtraction into $y - (-2) = y + 2$:

$$(7 - 3)^2 + (1 + 2)^2 = 4^2 + 3^2 = 16 + 9 = 25$$

Since $25 = r^2 = 5^2$, the point is exactly 5 units from the center, so yes, it lies on the circle.

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Common wrong answer 101: the center coordinates were added instead of subtracted, giving $(7 + 3)^2 + (1 - 2)^2$. The sign inside the bracket is the opposite of the sign of the coordinate — a center of $(3, -2)$ produces $(x - 3)$ and $(y + 2)$.

2. On $x^2 + y^2 = 25$, find both y when $x = 3$.

Substitute and solve; a circle is not a function, so a vertical line at $x = 3$ meets it twice and both roots are real answers.

$$\begin{aligned} 3^2 + y^2 &= 25 \\ y^2 &= 16 \\ y &= \pm 4 \end{aligned}$$

$$\boxed{y = -4 \text{ or } y = 4}$$

Grading note: reporting only $y = 4$ loses the point $(3, -4)$. The $\pm$ is not decoration; it is the second intersection.

3. Parabola with focus $(0, 3)$ and directrix $y = -3$.

The focus sits $p = 3$ above the vertex and the directrix sits 3 below it, so the standard form $x^2 = 4py$ becomes $x^2 = 12y$. Solve for y and substitute $x = 6$:

$$y = \frac{x^2}{4p} = \frac{6^2}{12} = \frac{36}{12} = 3$$

$$\boxed{3}$$

Common wrong answer 12: the equation was written as $x^2 = py$, so the student divided by 3 instead of by 12. The focal distance is p ; the coefficient in the equation is $4p$.

4. On $x^2 = 12y$, find both x when $y = 12$.

$$\begin{aligned} x^2 &= 12(12) = 144 \\ x &= \pm 12 \end{aligned}$$

$$\boxed{x = -12 \text{ or } x = 12}$$

A parabola opening upward is symmetric about the y -axis, so a horizontal line above the vertex always meets it twice.

5. Line $y = x + 1$ meets circle $x^2 + y^2 = 25$.

The line is already solved for y , so substitute it into the circle — one equation in one variable is the whole point of substitution:

$$\begin{aligned} x^2 + (x + 1)^2 &= 25 \\ 2x^2 + 2x + 1 &= 25 \\ 2x^2 + 2x - 24 &= 0 \\ x^2 + x - 12 &= 0 \\ (x - 3)(x + 4) &= 0 \end{aligned}$$

So $x = 3$ or $x = -4$. Substitute each back into the *line* (the easier equation): $y = 3 + 1 = 4$ and $y = -4 + 1 = -3$.

$$\boxed{(3, 4) \text{ and } (-4, -3)}$$

Check: $9 + 16 = 25$ ✓ and $16 + 9 = 25$ ✓

6. Complete the square: $x^2 + y^2 - 6x + 8y - 11 = 0$.

Group each variable and move the constant across:

$$(x^2 - 6x) + (y^2 + 8y) = 11$$

Half of -6 is -3 , and $(-3)^2 = 9$; half of 8 is 4 , and $4^2 = 16$. Add both to *each* side:

$$(x^2 - 6x + 9) + (y^2 + 8y + 16) = 11 + 9 + 16$$

$$(x - 3)^2 + (y + 4)^2 = 36$$

Read the center off the brackets (opposite signs) and the radius as $\sqrt{36}$:

center $(3, -4)$, $r = 6$

Common wrong answer $r \approx 3.32$: the constant 11 was taken as r^2 and the two added constants were never carried to the right-hand side. Whatever you add on the left must appear on the right, and here that turns 11 into 36 .

7. Circle $x^2 + y^2 = 25$ **meets parabola** $y = x^2 - 5$.

The parabola gives $x^2 = y + 5$, and substituting that into the circle eliminates x entirely — a much cleaner substitution than solving for x :

$$(y + 5) + y^2 = 25$$

$$y^2 + y - 20 = 0$$

$$(y + 5)(y - 4) = 0$$

So $y = 4$ or $y = -5$.

If $y = 4$: $x^2 = 4 + 5 = 9$, so $x = \pm 3$.

If $y = -5$: $x^2 = -5 + 5 = 0$, so $x = 0$ only — the parabola's vertex touches the bottom of the circle.

$(3, 4)$, $(-3, 4)$ and $(0, -5)$

Why three and not four: each y usually gives two x values, but $x^2 = 0$ gives a repeated root, so that intersection is a single tangential point.

8. Line $y = 2x - 3$ **meets parabola** $y = x^2 - 2x + 1$.

Both are solved for y , so set them equal:

$$x^2 - 2x + 1 = 2x - 3$$

$$x^2 - 4x + 4 = 0$$

$$(x - 2)^2 = 0$$

The discriminant is $16 - 16 = 0$: one repeated root, $x = 2$. Then $y = 2(2) - 3 = 1$.

$(2, 1)$

What the count tells you: a repeated root means the line touches the parabola at exactly one point without crossing it — the line is *tangent* at $(2, 1)$. Two roots would mean a secant, and no real roots would mean the line misses the curve entirely.

9. Locus equidistant from $(0, 5)$ and the line $y = -5$.

(a) Model derivation. The distance from $P(x, y)$ to the point is $\sqrt{x^2 + (y - 5)^2}$; the distance to the horizontal line $y = -5$ is the vertical gap $y + 5$. Setting them equal and squaring:

$$\begin{aligned}x^2 + (y - 5)^2 &= (y + 5)^2 \\x^2 + y^2 - 10y + 25 &= y^2 + 10y + 25 \\x^2 &= 20y\end{aligned}$$

Every y^2 and every constant cancels — that cancellation is what makes the locus a parabola rather than a circle. Comparing with $x^2 = 4py$ gives $p = 5$, which matches the focus $(0, 5)$ and directrix $y = -5$.

(b) Set $y = 5$: $x^2 = 20(5) = 100$, so $x = \pm 10$, and the points are $(-10, 5)$ and $(10, 5)$.

$x = -10 \text{ or } x = 10$

Manual review: part (a) is a written derivation, read by a teacher; part (b) is machine-checked. Credit (a) when the squaring step is shown and the y^2 terms are seen to cancel.

10. Circle $x^2 + y^2 = 25$ meets hyperbola $xy = 12$.

From the hyperbola, $y = \frac{12}{x}$ (and $x \neq 0$). Substitute:

$$\begin{aligned}x^2 + \frac{144}{x^2} &= 25 \\x^4 - 25x^2 + 144 &= 0 \\(x^2 - 9)(x^2 - 16) &= 0\end{aligned}$$

Treat it as a quadratic in x^2 : $x^2 = 9$ or $x^2 = 16$, so $x = \pm 3$ or $x = \pm 4$. Each x gives one $y = 12/x$:

$$x = 3 \Rightarrow y = 4; \quad x = 4 \Rightarrow y = 3; \quad x = -3 \Rightarrow y = -4; \quad x = -4 \Rightarrow y = -3.$$

$(3, 4), (4, 3), (-3, -4), (-4, -3)$

Sanity check: $xy = 12$ is positive, so x and y share a sign — no intersection can lie in the second or fourth quadrant, and none of the four does. Multiplying by x^2 was safe because $x = 0$ is not on the hyperbola.

11. Locus: $PA = 2 \cdot PB$ with $A(6, 0)$ and $B(0, 0)$.

Write both distances, then square immediately — the factor of 2 must be squared as well, which is where this problem is usually lost:

$$\begin{aligned}\sqrt{(x-6)^2 + y^2} &= 2\sqrt{x^2 + y^2} \\ (x-6)^2 + y^2 &= 4(x^2 + y^2) \\ x^2 - 12x + 36 + y^2 &= 4x^2 + 4y^2 \\ 0 &= 3x^2 + 3y^2 + 12x - 36 \\ x^2 + y^2 + 4x - 12 &= 0\end{aligned}$$

Complete the square in x : half of 4 is 2 and $2^2 = 4$, so

$$(x+2)^2 + y^2 = 12 + 4 = 16$$

center $(-2, 0)$, $r = 4$

Worth naming: “twice as far from A as from B ” produces a circle, not a line — the perpendicular bisector only appears when the ratio is 1. This family is the Apollonius circle. Sanity check with a point on the x -axis: $P(2, 0)$ gives $PA = 4$ and $PB = 2$ ✓.

12. Synthesis — dish $x^2 = 8y$ and ring $x^2 + y^2 = 20$.

(a) The dish equation already isolates x^2 , so substitute it into the ring:

$$\begin{aligned}8y + y^2 &= 20 \\ y^2 + 8y - 20 &= 0 \\ (y+10)(y-2) &= 0\end{aligned}$$

So $y = 2$ or $y = -10$. Only $y = 2$ is usable: $y = -10$ gives $x^2 = 8(-10) = -80$, which has no real solution — the parabola never reaches below its vertex at the origin. With $y = 2$: $x^2 = 16$, so $x = \pm 4$.

$(4, 2)$ and $(-4, 2)$

(b) Take $(4, 2)$. Distance to the focus $(0, 2)$:

$$\sqrt{(4-0)^2 + (2-2)^2} = \sqrt{16} = 4$$

Distance to the directrix $y = -2$ is the vertical gap $2 - (-2) = 4$.

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(c) Model answer: the two distances are equal, and that is not a coincidence about this one point — it is the definition of the parabola. Every point of the dish is exactly as far from the focus $(0, 2)$ as it is from the directrix $y = -2$. That property is why a dish shaped this way sends every signal arriving parallel to its axis to the same focus point.

Manual review: part (c) is written reasoning, read by a teacher; parts (a) and (b) are machine-checked. Credit (c) when the student states the equal-distance property as holding for *every* point of the curve.